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Life cycle assessment of shared electric bicycle on greenhouse gas emissions in China

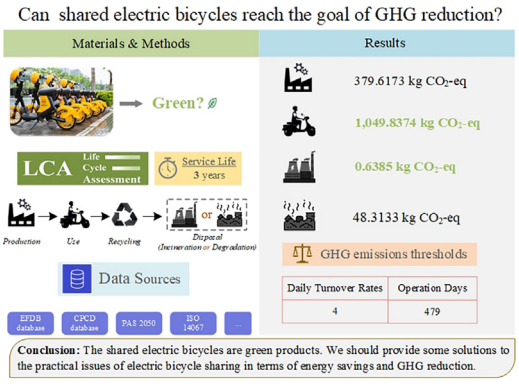
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HIGHLIGHTS

- Shared electric bicycles are green products under standard operating conditions.
- Shared electric bicycles can replace eleven kinds of transportation modes.
- GHG emissions are evaluated at all stages of the life cycle.
- Net GHG emission reductions are determined by different disposal methods.
- GHG emission thresholds are important to reach zero emissions.

GRAPHICAL ABSTRACT



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ABSTRACT

Following the bike-sharing system, the shared electric bicycle (SEB) is experiencing explosive growth in China as an emerging shared transportation mode. While shared transportation has long been linked to energy conservation and reducing emissions, a major problem facing SEB is whether it can reach the goal of greenhouse gas (GHG) reduction. This paper aims to evaluate GHG emissions at each stage of production, operation, and disposal of the SEB using life cycle assessment of GHG emissions. We also compared the differences in GHG emissions between the recycling incineration scenario (H1) and the recycling degradation scenario (H2) in the disposal stage. The GHG emissions of SEB in the production and operation stages were found to be 379.6173 kg CO_2-eq and 183.4663 kg CO_2-eq, respectively. However, the GHG emission reduction in the use stage was 1049.8374 kg CO_2-eq. Thus, the net GHG reduction in the life cycle of the SEB was 487.3923 kg CO_2-eq. (H1) and 433.9215 kg CO_2-eq. (H2), respectively, indicating that SEB had a green effect. Non-recyclable parts of SEB will take 48 years to degrade in landfills in the H2 scenario. The GHG emission thresholds for SEB were further discussed. When the average daily turnover rate of SEB was less than 4 and its operation day was less than 479, SEB would not be able to achieve the goal of GHG reduction in the whole life cycle, with the riding statistics remaining unchanged. Finally, some advice for practical issues of electric bicycle sharing in energy conservation and GHG reduction were presented in response to the results.

1. Introduction

Transportation is an important source of energy consumption and greenhouse gas (GHG) emissions and has become an important area for mitigating urban environmental pollution and promoting sustainable development. The transportation sector needs to achieve low-carbon

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development transformation as soon as possible in the context of a global response to climate change (Wang and He, 2018; Yuan et al., 2021). Shared transportation is a green development model innovation that supports sustainable urban transformation (Enochsson et al., 2021; Mi and Coffman, 2019). In recent years, shared electric bicycles (SEBs) have rapidly spread in many cities as an emerging green travel mode in China because of their advantage in speed, trip distance, and so forth (Chen et al., 2020a; Cherry et al., 2016). The number of SEBs increased significantly in recent years, from 175,000 to 1,157,000 between 2017 and 2020 in China with a compound annual growth rate of 105.4 %.

Electric bicycle sharing systems have become one of the fast-growing transportation modes in China and around the world including in Asia, North America, and Europe (He et al., 2019). Numerous studies related to electric bicycle sharing in these regions have been conducted by researchers, such as the first SEB in North America (Langford et al., 2013), user demand and system availability of SEB (Ji et al., 2014), factors influencing the choice of SEB (Campbell et al., 2016), health impacts and safety issues of electric bicycles (Guo et al., 2017; Otero et al., 2018), and more. However, the production, operation, and disposal of SEBs generate GHG emissions from a whole life cycle perspective. The result brings the question, "Can the SEB reach the goal of GHG reduction?" The question is inevitable and needs to be addressed in the electric bicycle-sharing systems for the future.

The modalities of shared mobility are often seen as promising options for sustainable transport. Examples include shared car, shared bicycle, and SEB. Many researchers have analyzed the various environmental impacts of car sharing, such as the impacts of car sharing on net GHG emissions as a result of changes in travel behavior (Shaheen et al., 2020; Martin and Shaheen, 2008), the life cycle impacts of car sharing on energy use and GHG emissions (Amatuni et al., 2020; Chen and Kockelman, 2016), the impact of mobility changes from car sharing on GHG reduction (Baptista et al., 2014), and more. Car sharing has been reported to reduce car ownership and personal annual vehicle kilometers, and promote a shift to other modes of transportation, with the potential to reduce GHG emissions (Arbeláez Vélez and Plepys, 2021; Namazu and Dowlatabadi, 2018). The emergence of shared bicycles provides new ideas for reducing GHG emissions and promoting urban sustainable travel (De Chardon, 2019; Ma et al., 2018). Bicycle sharing reduces CO_2 emissions compared to the previous mode of transportation used by users (Luo et al., 2019; D'Almeida et al., 2021). The bicycle-sharing program could improve air quality and increase sports activity rates (Anderson, 2015). Also, space, time, and the modal shift of different transportation modes have important effects on the environmental benefits of shared bicycles (Zhang and Mi, 2018; Wang et al., 2020). There are few studies on GHG emissions from SEB. Evaluating the environmental benefits of SEB in the use phase is the main direction, such as the cost and carbon benefits of the hybrid electric bicycle share system (Wamburu et al., 2021), the impact of SEBs and electric bicycles on achieving urban climate goals (Elliot et al., 2018; Felipe-Falgas et al., 2022; McQueen et al., 2020), and more. These studies show that SEBs have the potential to help cities and regions achieve climate goals. On the contrary, some research results indicated that SEBs have a substitution effect on other transportation modes and emitted more CO_2 than the transportation mode being substituted (Reck et al., 2022).

Various tools, such as environmental impact assessment (Ivanova et al., 2016) and strategic environmental assessment (Zhang et al., 2021), can be used to examine environmental implications of various systems. MFA and IO are two of the most common quantitative approaches, such as projections of economic benefits and GHG emissions of recycling systems (Liu et al., 2020; Tam et al., 2018), assessment of the ecosystem impacts of water scarcity (Feng et al., 2014), and more. As another prominent method, the Life Cycle Assessment (LCA) model focuses on products and supply chains from cradle to grave (Bishop et al., 2021). The different environmental impacts at various stages such as raw material extraction, production, use, and final recycling can be analyzed based on a large amount of data in the life cycle by the LCA model (Hellweg and Mila I Canals, 2014). The LCA model was used to assess the environmental impact of bicycles including classic bicycles (Chen et al., 2020b), shared bicycles (Coelho and Almeida, 2015), and electric bicycles (Li et al., 2014) to discover the main factors contributing to negative impacts.

The whole life cycle of SEB includes various stages, such as manufacture, use, waste disposal, and daily management. All stages of life cycle of SEBs require resource and energy inputs. It would generate waste and GHG emissions and indicate the potentially huge negative environmental impacts (Mao et al., 2021). Based on literature review, on the one hand, we discovered that none of these studies assessed the quantity of GHG emissions produced by SEB in terms of manufacture, operation, maintenance, and disposal over the life cycle. It may cause misconceptions about environmental effects of SEB. On the other hand, the key contributors to negative environmental impacts were not identified in these studies, which is critical to sustainable development of SEB. Moreover, the impact of SEB on energy savings and GHG reduction has not been fully included in the accounting of resource consumption.

The importance of SEB in energy conservation and GHG reduction should be discussed. Compared to other studies, the life cycle GHG emission accounting method is used to calculate GHG emissions from the production, operation, and disposal of the SEB. Data collection at all stages is the biggest difficulty in LCA modeling. These data cover the stage of production, management, and recycling. A variety of data sources are consulted, such as field survey data, consultancy studies, and literature. Therefore, both the theory and the data in this paper are sound. The net GHG reduction and the GHG emission thresholds can be calculated in the life cycle by investigating the use of the SEB. The results can be expected to provide emission factors for SEB, which are necessary information to support system analysis of urban transportation sustainability. Finally, several policies and governance proposals will be provided for the production design, market management, and recycling of the SEB to achieve energy savings and GHG reduction goals, which may be beneficial in improving the environmental contributions of SEB.

2. Materials and methods

The LCA analysis is carried out in accordance with ISO 14067:2018 requirements and methodology to determine GHG emissions throughout the life cycle of a product, including raw material acquisition, material production, manufacturing, assembly, use, and disposal (reuse, recycling, incineration, or landfill). The four steps of LCA, as defined by ISO rules, are Goal and Scope Definition, Inventory Analysis, Impact Assessment, and Results Interpretation (ISO, 1997). The first step in the LCA is the Goal and Scope Definition, which clarifies the questions to be answered in the LCA. It controls the scope and depth of the research. The next important phase in LCA is Life Cycle Inventory (LCI) Analysis, which collects, compiles, and quantifies the inputs and outputs of the research object (Lu et al., 2020). The daily transit, recycling, use stage, and every stage of manufacture are all tracked and built into the inventory data using the Inventory Analysis. The Impact Assessment is the third step of the LCA, which is inseparable from LCI. The Impact Assessment evaluates environmental impact of each stage using a database according to the Inventory Analysis results. It will provide information for the next stage (the Results Interpretation). The Results Interpretation is the last step of LCA, under the content of the Goal and Scope Definition. It identifies the key issues that have a significant environmental impact by conducting a thorough examination of the findings from the Life Cycle Inventory and the Impact Assessment. The limitations of this study and useful advice were also discussed.

2.1. Definition of goals and scope

The LCA model and PAS 2050 GHG emissions accounting methods are used to analyze the GHG emissions of the SEB in China. The environmental impact of SEB is assessed from a life cycle perspective. The entire life cycle includes the extraction of resources, preparation of the raw materials, product assembly, shipping, use, upkeep, and recovery of waste.

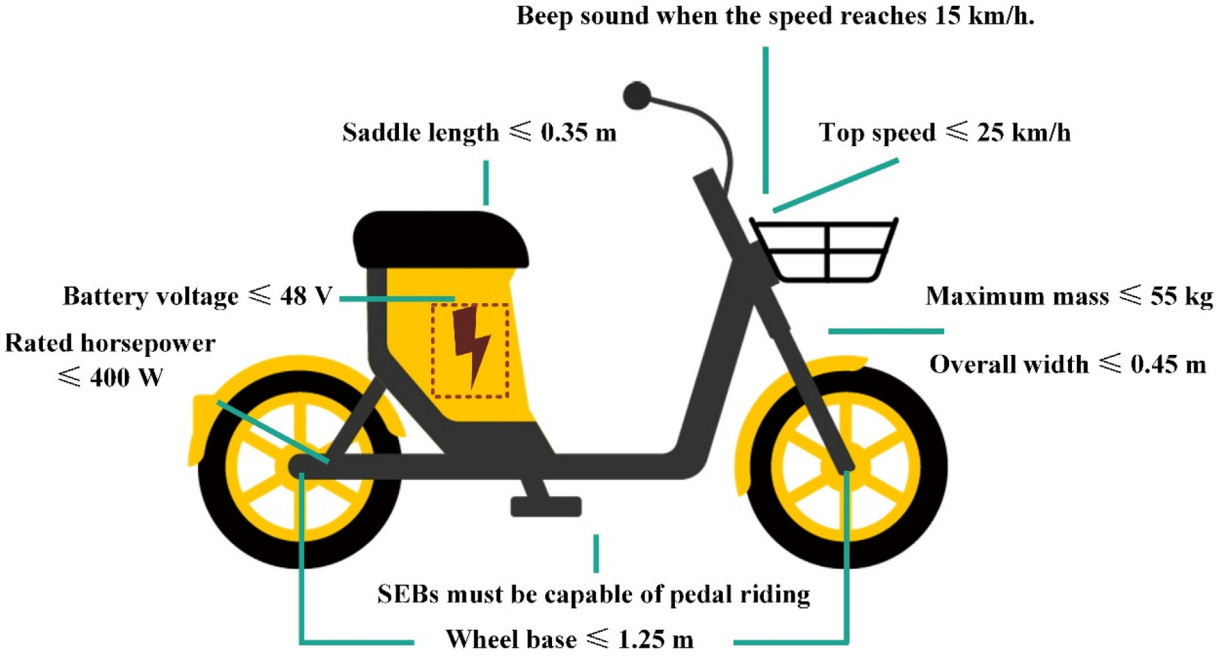


Fig. 1. The main technical specification for the electric bicycle (GB17761–2018).

Electric bicycle sharing programs, such as those in Riverside, Raleigh, and Guildford, have gradually become popular around the world in recent years. Since 2017, there have been more than ten kinds of SEBs related to eight brands on the market, such as Meituan, Qingju, Hello, Songguo and Xiaoliu. Bicycle-sharing giants Meituan and Qingju, both of which use lead-acid batteries, accounted for nearly 63.3 % of the market share in China (Leadleo, 2021). Therefore, the LCA model of electric bicycle sharing system is established based on the operation of Meituan SEB, which includes user data from 256 cities in China.

2.1.1. The characteristics of electric bicycle sharing

SEBs are now entering people's lives as a new shared transportation mode for the 3–10 km travel market due to the wide application of new technologies such as the Internet of Things (IoT), Artificial Intelligence (AI), and Big Data. There are great differences in different cities in terms of market access standards and quantity limits for the SEB.

Before 2018, there was a lack of unified product standards for electric bicycles. Some metropolises in China, such as Beijing and Shanghai discouraged the development of the SEB due to concerns about travel safety, city image, and traffic congestion. With the implementation of the Safety Technical Specification for Electric Bicycle (GB17761-2018) in 2019 (Fig. 1), there are new development opportunities for SEB in China. The users of SEB are mainly in third-tier and fourth-tier cities because of the influence of policies and technologies. Compared with other traffic tools, it was generally believed that the SEB was safer for users. 58.7 % of users hold a positive attitude towards the future development of SEB (Iimedia, 2020).

2.1.2. Functional unit and system boundary

The life cycle analysis approach is used to examine key aspects of energy consumption of SEB. The functional unit chosen is transporting one passenger for one kilometer by a SEB (i.e., p·km). The SEB service period is three years (SPIR, 2020). The entire life cycle of the SEB can be separated into three stages: production, use, and disposal. The steps are divided into four sections such as the production section, alternative section, operation section, and disposal section. Energy used to manufacture SEB occurs in the production section. The operation section consumes energy to dispatch and maintain SEB. The disposal section generates energy consumption for the disposal of SEB. All processes involve energy consumption. However, the SEB saves energy by replacing other transportation modes during the alternative section. It involves energy savings as shown in Fig. 2. The system boundary is the GHG generated by the input of each component into the assembly plant to SEB recycling, according to the data availability and continuity. It includes GHG emissions from materials, water, power, gas, and oil used in SEB manufacturing, transportation, maintenance, and recycling.

2.2. Impact assessment method

2.2.1. Applying GHG accounting methods to the production stage

The SEB production process consists of four major steps: raw material processing, frame spraying, assembly, and transportation. Energy consumption of raw material processing occupies the main part of all energy consumption in the production process. GHG emissions of various raw materials can be calculated using the "PAS 2050:2011 for the Evaluation of Greenhouse Gas Emissions of Goods and Services in the Life Cycle" standard[1] and the China Products Carbon Footprint Factors Database (CPCD).[2] The following is the formula for calculating GHG emissions in the production process:

$$PPE = \sum_{1}^{n} EC_n * EF_n \quad (1)$$

PPE: GHG emissions in the production process.EC_n: the *nth* energy consumption in the production process.EF_n: GHG emission factor for the *nth* energy consumed in the production process.

2.2.2. Applying GHG accounting methods to the use stage

According to the usage scenarios of SEB, SEB, as a new shared transportation mode, can replace other transportation modes including cars, buses, personal bicycles, etc. It might contribute to the reduction of GHG emissions. The GHG emission factors of the SEB can be obtained in the use stage according to the China Certified Emission Reduction Methodology (CCER).[3] The following is the formula for calculating in the use stage:

$$UPE = EF_{SEB} * DIS_{avg} \quad (2)$$

[1] https://www.doc88.com/p-28447071035020.html.
[2] http://lca.cityghg.com.
[3] https://www.ccer.com.cn.

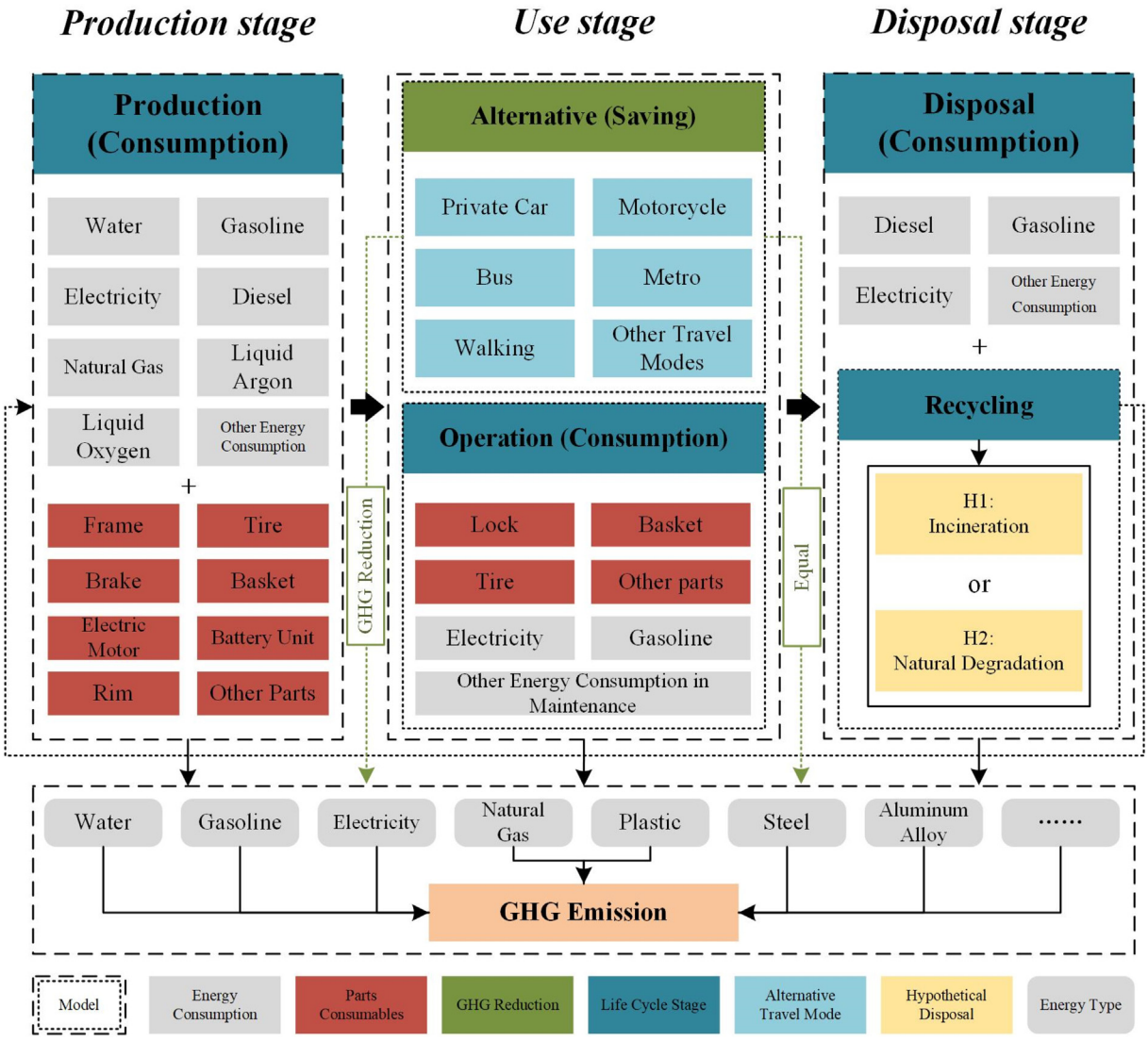


Fig. 2. Life cycle unit process and system boundary of the SEB.

$$EF_{SEB} = \sum_{i}^{n} (TR_i * TF_i) - 0.53H \tag{3}$$

UPE: GHG emissions in the use stage.EF_{SEB}: GHG emission factor of SEB. DIS_{avg}: average travel distance of SEB each time.TR_i: the proportion of users who travels with SEBs instead of other transportation modes *i*.TF_i: GHG emission factor of transportation mode *i*.*H*: average power consumption per kilometer of the SEB.

In 2020, the average GHG emission factor of the grid was 0.53 kg CO_2-eq/kW·h in China.[4].

2.2.3. Applying GHG accounting methods to the operation and dispatch stage

The operation and dispatch stage includes maintenance section and rebalancing section. The maintenance of SEB includes pumping, lubricating, and cleaning. Unfortunately, only the replacement of components during maintenance was considered due to the intricacy of these activities and the challenge of data gathering. All SEBs are distributed and rebalanced by vans among stations/locations in the rebalancing section. The following is the formula for calculating GHG emissions in the operation and dispatch stage:

$$OPE = RPE + MPE \tag{4}$$

$$RPE = EF_{day} * T_l \tag{5}$$

$$MPE = \sum_{1}^{m} EC_m * EF_m * ME_m \tag{6}$$

OPE: GHG emissions in the operation and dispatch stage.*RPE*: GHG emissions in the rebalancing section.*MPE*: GHG emissions in the maintenance section.EF_{day}: daily average GHG emissions of SEB in the rebalancing section.T_l: the time length of life cycle of SEB, which is three years.ME_m: repair probability of the *mth* component.

2.2.4. Applying GHG accounting methods to the disposal stage

The recyclable parts of SEB are recycled by renewable resource recycling enterprise in the disposal stage. There are two different disposal methods for the non-recyclable parts of the SEB, including incineration and natural degradation. As a result, two disposal scenarios are constructed based on distinct disposal methods. GHG emissions of SEB in the disposal stage are calculated based on the actual disposal scenario.

[4] https://lca.cityghg.com.

Scenario 1. SEB is divided into recyclable and non-recyclable components. The non-recyclable components are incinerated as municipal solid waste once the recyclable components have been recycled. According to some research and interviews, the recyclable components of the SEB are primarily the frames, electric motors, battery units, and other metal parts. According to the calculation formula for GHG emissions from MSW incineration, GHG emissions are calculated as:

$$DPE_{H1} = \sum_{i}^{n} MSW_i * EF_i \quad (7)$$

DPE_{H1}: GHG emissions of scenario 1.MSW_i: weight of municipal solid waste of component type *i*.EF_i: GHG emission factor for the incineration of type *i* component.

Scenario 2. The non-recyclable components of the SEB are piled up, dumped, and left to degrade naturally in landfill sites. The disposal method takes into account the dispatch and GHG emissions of heavy machinery such as cranes, forklifts, and tractors. The material properties of each component in the natural deterioration stage are used to calculate different GHG emissions. The primary component of tires is rubber, which is a high-strength, aging-resistant polymer material. It takes hundreds of years for tires to decompose enough to not affect the environment. The degradation of plastics mainly includes biodegradation, photolysis, and chemical degradation. It is affected by temperature and humidity. This process will generate a lot of GHG, such as CH_4 and CO_2.

The following is the formula for calculating GHG emissions of scenario 2:

$$DPE_{H2} = \sum_{j}^{n} MSW_j * EF_j \quad (8)$$

DPE_{H2}: GHG emissions of scenario 2.MSW_j: weight of the *jth* non-recyclable part.EF_j: GHG emission factor of the *jth* non-recyclable part.

The maximum design life of landfill site is ten years according to the Harmlessness Evaluation Standard of Domestic Garbage Landfill (CJJ/T107-2019) released by the MOHURD of China. According to the time length of the life cycle, it was assumed that the practical stage emissions would occur in three years. GHG would be gradually released into the atmosphere over the following ten years. The weighted average time factor is calculated as follows:

$$NDWT_c = \frac{\sum_{i=1}^{100} X_i(100 - i)}{100} \quad (9)$$

$NDWT_c$: the weighted coefficient of natural degradation time.X_i: the fraction of emissions in the *ith* year to overall emissions, which is set to 0.1.*i*: the *ith* year of emissions.

According to the formula, $NDWT_c$ is calculated to be 0.915.

$$T_{ET} = DPE_{H2} * NDWT_c \quad (10)$$

T_{ET}: the average existing time for the total GHG emissions produced by SEB in the atmosphere.

It can be calculated according to the formula over the 100-year evaluation period.

2.2.5. The net GHG reduction of the whole life cycle of SEB

After the above calculations, GHG emissions at each stage of SEB life cycle are summed up. Finally, the net GHG emissions of the whole life cycle of SEB can be calculated:

$$LPE_{Hi} = PPE + UPE + OPE + DPE_{Hi} \quad (11)$$

LPE_{Hi}: net GHG emissions during the entire life cycle of the SEB.

Table 1
Summary of the main emission factors used to calculate GHG emissions of SEB.

Coding	Material	GHG emission factor	Unit	Reference
1	Steel	2.3000	t CO_2-eq/t	CPCD database
2	Tire	2.7300	t CO_2-eq/t	CPCD database
3	Aluminum alloy	18.3000	t CO_2-eq/t	CPCD database
4	Gasoline	0.2850	kg CO_2/km	EFDB database[a]
5	Diesel	0.6026	kg CO_2-eq/(t·km)	CPCD database
6	Paper	0.1400	t CO_2-eq/t	CPCD database
7	Lead-acid battery	0.1100	kg CO_2-eq/kg battery	CPCD database
8	Municipal solid waste incineration	−124.3000	kg CO_2-eq/t	CPCD database
9	Natural degradation	15.8000	% (by weight)	EFDB database
10	Natural degradation-Plastic	0.1300	kg CH_4/kg	EFDB database
11	Natural degradation-Rubber	0.4700	kg CH_4/kg	EFDB database
12	Natural degradation-Paper	0.4400	kg CH_4/kg	EFDB database
13	Plastic	0.1000	% (by weight)	EFDB database
14	Electricity	0.5300	kg CO_2/kW·h	GB/T 32151.1
15	Water	0.9110	kg CO_2/m^3	Carbonstop[b]
16	Natural gas	2.0000	kg CO_2/m^3	Carbonstop
17	Sewage treatment	0.2400	kg CO_2/m^3	(Hirano and Fujita, 2016)

[a] https://www.ipcc-nggip.iges.or.jp/EFDB/main.php.
[b] https://www.carbonstop.net.

It is determined by which disposal method is taken for the non-recyclable components at the disposal stage.

3. Life cycle inventory analysis

3.1. Determining the types and sources of GHG emission factors

The object of this research was the electric bicycle-sharing industry in China, represented by Meituan SEB. The basic parameters in the calculation were determined using the above GHG calculation methods. GHG accounting was performed at three stages during the life cycle. Finally, GHG conservation thresholds were obtained. Table 1 lists the types and sources of vehicle GHG emission factors.

3.2. GHG emissions in the production stage

SEB companies typically contract out production to conventional electric bicycle factories. The SEB can be treated as an ordinary electric bicycle. Electric bicycle components manufacturing primarily comprises creating all spare parts and auxiliary materials. Two components make up the majority of the SEB: the bicycle system and the technical product system (such as chips). The technical product system is often light in weight and recyclable. Therefore, only the material and energy usage of the SEB is taken into account at the production stage.

Material and energy usage in the SEB production stage was collected from electric bicycle manufacturers, distributors, and some research. Through electric bicycle manufacturers and distributors, the SEB weighed about 51.100 kg, including 13.522 kg of steel and copper, 17.310 kg of aluminum alloys, 3.210 kg of plastic, 3.500 kg of rubber, 11.700 kg of lead alloys[5], 0.060 kg paper, etc. 0.0026 m^3 of water, 0.9639 kW·h of electricity, and 2.2143 m^3 of natural gas were all used to produce a SEB. Other auxiliary materials (such as lubricants, and paint manufacture) were utilized in significant amounts at the factory, but because they were involved in a SEB in small quantities, we could disregard this portion by the trade-off

[5] The lead content of electric bicycle batteries is about 85 %, and the lead consumption in production is about 1.02 times that of lead content.

Table 2
GHG emission calculation of the SEB production materials.

Coding	Part name	Material type	Weight (kg)	GHG emission factor	Unit	GHG emissions (kg)
1	Frame & frame parts	Aluminum alloy	13.0000	18.3000	t CO_2-eq/t	237.9000
2	Front fork	Aluminum alloy	0.9100	18.3000	t CO_2-eq/t	16.6530
3	Front fork components	Aluminum alloy	0.2800	18.3000	t CO_2-eq/t	5.1240
4	The horizontal	Steel	1.4000	2.3000	t CO_2-eq/t	3.2200
5	Handles	PVC	1.0000	3.4100	kg CO_2-eq/kg	3.4100
6	The brake#1	Steel	0.3920	2.3000	t CO_2-eq/t	0.9016
7	The brake#2	Aluminum alloy	0.1200	18.3000	t CO_2-eq/t	2.1960
8	Foot support	Steel	0.4000	2.3000	t CO_2-eq/t	0.9200
9	Sprocket crank	Steel	0.4500	2.3000	t CO_2-eq/t	1.0350
10	The chain	Steel	0.3000	2.3000	t CO_2-eq/t	0.6900
11	The Pedal	Plastic	0.3300	3.1200	kg CO_2-eq/kg	1.0296
12	Rims	Aluminum alloy	3.0000	18.3000	t CO_2-eq/t	54.9000
13	Tires	Rubber	2.0000	2.7300	t CO_2-eq/t	5.4600
14	Basket components	Steel	0.5000	2.3000	t CO_2-eq/t	1.1500
15	Lock	Steel	0.7800	2.3000	t CO_2-eq/t	1.7940
16	Lighting sets	Plastic	0.2000	3.1200	kg CO_2-eq/kg	0.6240
17	The saddle	Rubber	1.5000	2.7300	t CO_2-eq/t	4.0950
18	Battery box	Plastic	0.1000	3.1200	kg CO_2-eq/kg	0.3120
19	Fender components	Steel	1.9000	2.3000	t CO_2-eq/t	4.3700
20	Applique	Paper	0.0600	0.1400	t CO_2-eq/t	0.0084
21	The sign	Plastic	0.0800	3.1200	kg CO_2-eq/kg	0.2496
22	Suspension parts	Steel	2.4000	2.3000	t CO_2-eq/t	5.5200
23	Guard plates	Plastic	1.5000	3.1200	kg CO_2-eq/kg	4.6800
24	Electric motor	Steel & copper	5.0000	16.8200	kg CO_2-eq/unit	16.8200
25	Battery unit	Lead-acid battery	13.5000	0.1100	kg CO_2-eq/kg battery	1.4850
	Total		51.1020			374.5472

concept. Meituan outsourced the production of SEBs to SUNRA, which has production bases in Wuxi, Xiangyang, and Tianjin. At present, Meituan SEBs have been mainly concentrated in third-tier and lower-tier cities, accounting for 73.9 % of the total number of SEBs launched in 2020 (Leadleo, 2021). According to the distance measurement of the map software, the average transportation distance (road transportation) of 100 SEBs from the production base to third-tier and third-tier cities was about 250.58 km. Therefore, the average transportation distance of the SEB was about 2.51 km, producing 0.1283 kg CO_2-eq. Table 2 summarizes the materials used during the production process as well as their GHG emissions. Finally, the total GHG emissions of SEB in the production stage is calculated to be 379.6173 kg CO_2-eq.

3.3. GHG emissions in the use stage

The data in the use stage of SEB included energy consumption data obtained from Meituan and the CPCD database. User behavior data was provided by a questionnaire survey that covered 256 cities in China. The average distance SEB traveled per trip was 3.06 km. The power consumption per kilometer was 0.014 kW·h. The daily turnover rate was between six and ten. According to the Technical Requirements for Chargers for Electric Bicycles (GBT36944-2018)[6] released by the SAMR of China, the actual power consumption per kilometer of the SEB is 0.0169 kW·h due to the loss in the charging process[7]. It was assumed that the daily turnover rate of the SEB was eight. Therefore, the average daily distance of the SEB was 24.48 km. We found that SEB could replace 11 kinds of transportation modes including buses, cars, metros, etc. from the questionnaire. Fig. 3 shows that buses are the transportation modes with the largest proportion replaced by the SEB. SEB mainly replaced the transportation modes that produced GHG, indicating that it had the effect of reducing GHG emissions. Fig. 4 shows the GHG emission factors of different transportation modes in the use stage. According to the calculation method of GHG emissions, when the SEB traveled 1 km, it would reduce 39.2 g of GHG emissions, showing that SEB was a green product. Finally, we could calculate that total GHG reductions during the use stage of SEB were 1049.8374 kg CO_2-eq.

3.4. GHG emissions in the operation and dispatch stage

Energy consumption data in the maintenance stage of the SEB was collected from interviews with the staff of Meituan SEB. Table 3 shows the failure rates of SEB components in one month. The failure rate of brakes, tires, baskets, and other components of SEBs is relatively high in Table 3. As the service time went by, SEB would require approximately seven maintenance over the lifetime. After the 7th restoration, the severity of damage to SEB would be considerably worsened, and usually beyond repair. It was calculated that the GHG emissions in the maintenance stage of the SEB were 5.5288 kg CO_2-eq in the whole life cycle according to the calculation method of GHG emissions.

It is a big issue when SEBs park illegally. SEB companies employ dispatch vans to transport and redistribute SEBs to maintain a good city image. The dispatch and redistributing of SEBs include SEB handling, placing, and battery replacement. It mainly relies on manual handling and transportation, which is divided by region and carried out in 24-h shifts. Small trucks and electric three-wheelers are the primary dispatch vehicles, which are similar to the general cargo transport system. Therefore, the focus was on GHG emissions from interval dispatching and peripheral reflux. GHG emission factors of these vehicle types were 0.083 kg CO_2-eq/(t·km) and 0.081 kg CO_2-eq/km, respectively, with a reference to Carbonstop and CPCD. In the dispatch stage, the staff of Meituan used small trucks to deliver the stacked SEBs to other parking stations that lack SEBs, and at the same time recover the broken SEBs to ensure the safety of users. One worker could dispatch 120 SEBs a day. Every dispatching took about 1 h. The small truck transported SEBs around the work area about 6 times a day. The staff mainly used a three-wheeled electric vehicle to carry batteries with sufficient power in the battery replacement section. According to system message, they discovered the SEB with low battery power on the road and replaced the battery to ensure that the SEB can be used normally. The staff replaced the batteries from 8:00 to 18:00, from

[6] https://www.miit.gov.cn/cms_files/filemanager/oldfile/miit/n1146285/n1146352/n3054355/n3057497/n3057502/c6176772/part/6176777.pdf.

[7] According to the GBT36944-2018 (https://www.samr.gov.cn), electric bicycles should not be charged for more than 8 h. The VOUT (output voltage) of 48 V electric bicycle charger is between 55 V and 59.6 V, The IOUT (current output) is between 2.1 A and 3.0 A. After averaging VOUT and IOUT respectively, the 48 V electric bicycle charger consumes about 1.1689 kW·h once charged. The discharge capacity of standard battery (48 V 20 Ah) used by the SEB is about 0.96 kW·h. Therefore, the loss of charge of battery is about 21 % after charge. Finally, we can get the actual power consumption per kilometer of the SEB. Calculation process: $0.014\text{kW}\cdot\text{h} * 1.21 = 0.0169\text{kW}\cdot\text{h}$.

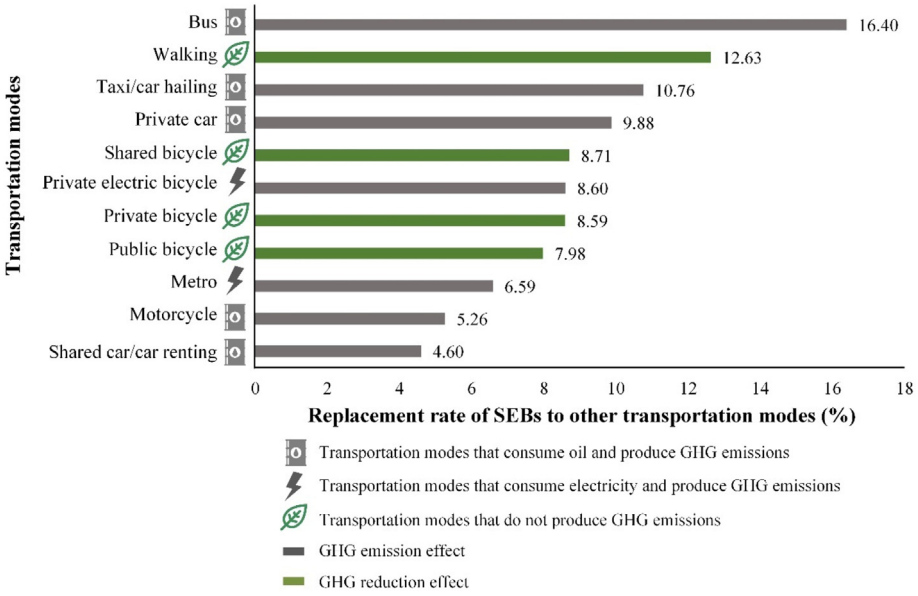


Fig. 3. the replacement rate of the SEB to other transportation modes.

22:00 to 6:00 the next day, with a 2-h break in between. One worker could replace 300 batteries a day. It was assumed that 25 % of SEBs were transported by dispatch every day. According to the Code for Design of Urban Road Traffic Facility (GB 50688-2011)[8] in China, the average road speed in class C and D cities is 30 km/h[9]. Finally, the average GHG emissions of the SEB were calculated at 0.1625 kg CO_2-eq/day in the dispatch stage. The data description of the operation and dispatch stage is shown in Appendix Table C.

3.5. GHG emissions in the disposal stage

Energy consumption data in the disposal stage were obtained from the EFBD database. The disposal consists of two scenarios: recycling incineration and recycling degradation. The recyclable parts of SEB were recycled by the renewable resource recycling enterprise, including frames, electric motors, battery units, and other metal parts. The non-recyclable parts mainly included 3.50 kg of rubber, 3.21 kg of plastic, and 0.06 kg of paper. We assumed that technical products, such as electronic chips, could be completely recycled and repurposed. According to the Principle of Landfill Site Selection (CJJ17-2004)[10], the distance between the landfill site and the urban planning built-up area should be greater than 5.00 km, and the distance from residential areas should be longer than 500 m. At present, there is no uniform regulation on the distance between waste incineration plants and renewable resource recycling plants in urban living areas. It will be ultimately determined by "environmental impact assessment" of the project.

Shenyang was taken as an observation area because of its good operation of SEB. There are three waste incineration plants in Shenyang and more than 110 renewable resource recycling plants. The map software was used to measure the distance of waste incineration plants from urban residential areas and the distance of renewable resource recycling plants from urban areas. The average of the distances was calculated to be 28.93 km and 1.00 km, respectively. Therefore, it was assumed that the distance from urban area to renewable resource recycling plant was 1.00 km, the distance from renewable resource recycling plant to waste incineration plant was 27.93 km, and the distance from the renewable resource recycling plant to the landfill was 4.00 km. 1) In the recycling incineration scenario, diesel vehicles could transport 100 SEBs from urban area to a renewable resource recycling plant 1.00 km away for recycling. Then, non-recyclable parts as municipal solid waste including rubber, plastic, and paper were transported to a waste incineration plant 27.93 km away for incineration. The average transportation distance of SEB was 0.047 km. It would reduce a total of 0.6385 kg CO_2-eq in the recycling incineration scenario according to the GHG emission calculation method. 2) In the recycling degradation scenario, non-recyclable parts were transported by diesel vehicles to a landfill site 4.00 km away. The average transportation distance of the SEB was 0.015 km. We found that the SEB would produce 52.8323 kg CO_2-eq of GHG emissions according to the GHG emission calculation method in Scenario 2. Table 4 lists the GHG emissions of the SEB in different scenarios of the disposal stage.

4. Results and sensitivity analysis

4.1. GHG emissions assessment results and sensitivity analysis

According to the calculation methods of GHG emissions at different stages of the life cycle, GHG emissions were calculated to be 379.6173 kg CO_2-eq in the production stage of SEB. It was the largest GHG emission in the life cycle. We calculated that GHG emissions in the operation and dispatch stage is 183.4663 kg CO_2-eq in the whole life cycle. SEB had a green effect due to the substitution of other transportation modes in the use stage. It would reduce GHG emissions by 1049.8374 kg CO_2-eq over the life cycle. GHG emissions from municipal solid waste piling and landfilling were much greater than those from incineration for electricity production in China owing to technical limitations. Therefore, there was a large gap in GHG emissions between recycling incineration and recycling degradation. GHG reductions of SEB were 0.6385 kg CO_2-eq in the recycling incineration scenario. GHG emissions of SEB were 52.8323 kg CO_2-eq in the recycling degradation scenario. Fig. 5 shows GHG emissions at different stages of the SEB life cycle.

Sensitivity refers to the rate of change of a certain LCA result caused by the unit rate of change of inventory data, which is equivalent to the contribution rate of each consumption or emission of all processes to the LCA results. We made the analyzed impact factors a 10 % reduction while keeping other impact factors constant, and then compared the changes in life cycle GHG emissions of SEB under the two scenarios. Table 5 shows the impact of each stage of the SEB life cycle on GHG emissions. As can be seen from Table 5 and Fig. 5, the impacts on GHG emissions in the SEB production

[8] https://www.mohurd.gov.cn/gongkai/fdzdgknr/tzgg/201908/20190829_241614.html.
[9] https://www.mohurd.gov.cn/gongkai/fdzdgknr/tzgg/201908/20190827_241566.html.
[10] https://www.mohurd.gov.cn/gongkai/fdzdgknr/tzgg/200402/20040226_155623.html.

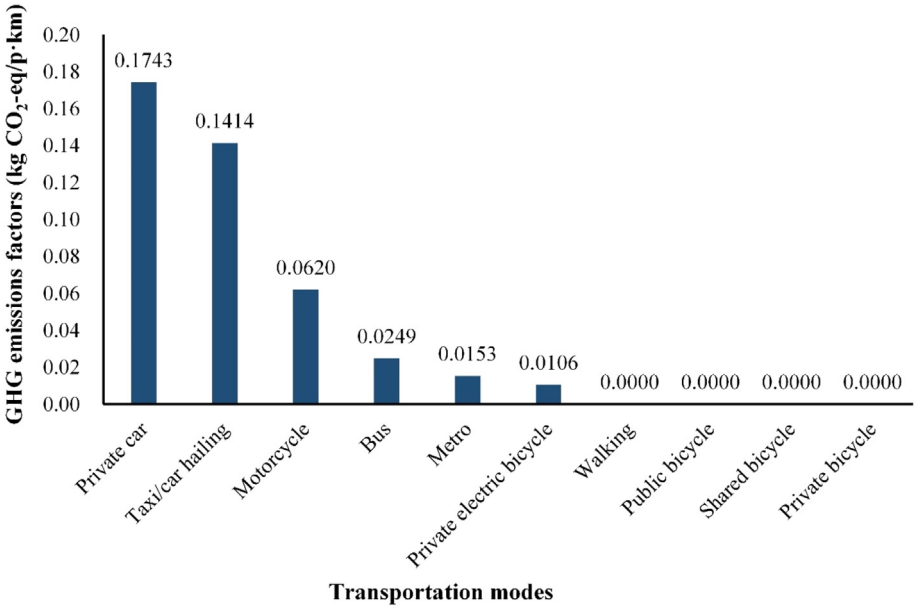


Fig. 4. the GHG emission factors of different transportation modes in the use stage.
Notes: We assumed that the GHG emissions of human-powered transportation modes are zero in the use stage, such as walking and cycling. Appendix Table A presents a reference to GHG emission factors of different transportation modes in the use stage.

Table 3
GHG emissions calculation of the SEB maintenance materials.

Coding	Part name	Maintenance probability (%)	Weight (kg)	GHG emission factor	Unit	GHG emissions (kg)
1	The brake#1	0.7000	0.3920	2.3000	t CO_2-eq/t	0.0063
2	The chain	2.2300	0.3000	2.3000	t CO_2-eq/t	0.0154
3	Foot support	2.9700	0.4000	2.3000	t CO_2-eq/t	0.0273
4	The saddle	4.4000	1.5000	2.7300	t CO_2-eq/t	0.1802
5	The brake#2	5.5700	0.1200	18.3000	t CO_2-eq/t	0.1223
6	The Pedal	5.6300	0.3300	3.1200	kg CO_2-eq/kg	0.0580
7	Lock	58.1300	0.7800	2.3000	t CO_2-eq/t	1.0429
8	The sign	61.7300	0.0800	3.1200	kg CO_2-eq/kg	0.1541
9	Tire	69.0300	2.0000	2.7300	t CO_2-eq/t	3.7690
10	Basket components	13.3300	0.5000	2.3000	t CO_2-eq/t	0.1533
	Total					5.5288

stage constitute the highest. The consumption of aluminum alloy, rubber, and plastic has the most significant impact on GHG emissions. Lower failure rates also help reduce GHG emissions during the operation and dispatch stage. The reduction in rubber, plastic, and paper leads to a reduction in MSW in the disposal stage, which increases GHG emissions in H1. The impact factors such as average trip distance, daily turnover rate, service period, and replacement rates for transportation are directly related to the GHG emission reduction in the use stage. The magnitude of changes in GHG emission reduction is consistent with the changes in the above impact factors according to formulas (2) and (3). The replacement rates for high-emission transportation by SEB have a significant impact on GHG reduction in the use stage, which is very critical.

4.2. Analysis of net GHG emission reductions from the SEB

If riding data remained constant, the overall GHG reduction factor for travel would stabilize at 0.0392 kg CO_2-eq/p·km. It implies that when a SEB traveled 1 km, it could reduce GHG emissions by 39.2 g. Regardless

Table 4
GHG emissions calculation of a SEB in the disposal stage.

Operation	Energy consumption	Weight	Unit	GHG emission factor	Unit	GHG (kg CO_2-eq/SEB)
Scenario 1	Solid waste incineration	6.7700	kg	−124.3000	kg CO_2-eq/t	0.8415*
	Diesel van average	0.0470	km	0.6026	kg CO_2-eq/(t·km)	0.1447
	Electricity	0.1100	kW·h	0.5300	kg CO_2-eq/kW·h	0.0583
	Total					0.6385*
Scenario 2	Sewage	0.0070	m^3	0.2400	kg CO_2/m^3	0.0064
	Diesel van average	0.0150	km	0.6026	kg CO_2-eq/(t·km)	0.0309
	Mechanical energy consumption	2.0000	km	0.2850	kg CO_2/km	0.5700
	Natural degradation-Plastic	3.2100	kg	0.1300	kg CH_4/kg	0.4176
	Natural degradation-Rubber	3.5000	kg	0.4700	kg CH_4/kg	1.6450
	Natural degradation-Paper	0.0600	kg	0.4400	kg CH_4/kg	0.0264
	Total					52.8323

Notes:
* Is for GHG reduction.

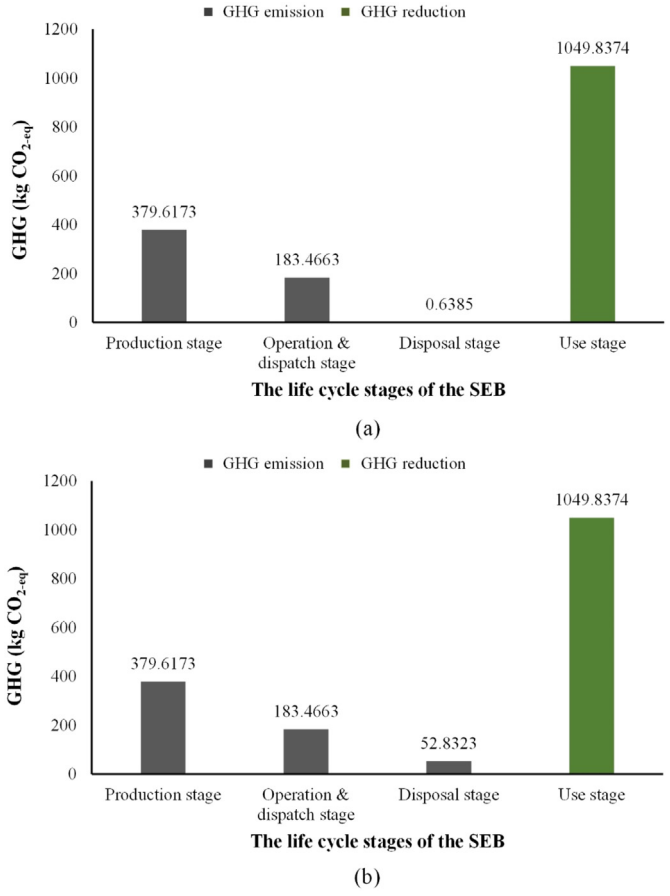


Fig. 5. GHG at different stages of the SEB life cycle. (a) In the recycling incineration scenario (H1); (b) In the recycling degradation scenario (H2).

of the disposal scenario, it could reduce GHG emissions by a total of 1049.8374 kg CO_2-eq over the life cycle. Therefore, the total net GHG emission reductions of the whole life cycle were 487.3923 kg CO_2-eq and 433.9215 kg CO_2-eq for the two disposal scenarios, respectively (Table 6). A well-operating SEB could reduce GHG emissions by 396 g - 445 g per day. Thus, the SEB had a significant green effect.

Table 5
Results of sensitivity analysis of SEB life cycle.

Various stages	Impact factors	Change in H1 (%)	Change in H2 (%)
Production stage	Aluminum alloy	5.6321	5.1431
	Steel	0.3485	0.3182
	Rubber	0.1699	0.1551
	Plastic	0.1226	0.1120
	PVC	0.0606	0.0554
	Lead alloy	0.0264	0.0241
	Paper	0.0001	0.0001
Use stage	Average trip distance	9.9192	9.9192
	Daily turnover rate	9.9192	9.9192
	Service period	9.9192	9.9192
	Replacement rates for low and zero-emission transportation	−3.0338	−3.0338
	Replacement rates for high-emission transportation	24.8381	24.8381
Operation & dispatch stage	Failure rate	0.0983	0.0898
Disposal stage	Rubber	−0.0077	0.0267
	Plastic	−0.0071	0.0068
	Paper	−0.0001	0.0004

Notes: According to the actual scenario, we analyzed the GHG emission reduction changes of impact factors in the use stage.

4.3. GHG emissions threshold analysis of the SEB

SEB had been proven to have a green effect of reducing GHG emissions. Fig. 6a shows it will take at least 479 days for SEB to meet net positive GHG emissions point in the recycling incineration scenario (scenario 1). 16 months are needed for the SEB to reach zero GHG emissions and green travel status. Similarly, Fig. 6b shows that it will take a minimum of 511 days to reach net positive GHG emissions point in the recycling degradation scenario (scenario 2). The SEB must be in service for at least 17 months to reach zero GHG emissions and green travel status.

The increasing number of SEBs would lead to a gradual increase in the rate of empty SEBs. We assumed that the increased availability of SEB would not improve people's inclination to ride them. Therefore, the daily turnover rate of SEB would gradually decrease. Fig. 7 shows that if the average daily turnover rate of the SEB is less than four times, the SEB will not meet the goal of reducing GHG emissions under the condition that the riding statistics remain unchanged in the whole life cycle.

4.4. Analysis of GHG reduction of electric bicycle sharing in China

The market research findings showed that the major concern of SEB users was the difficulty in finding and returning SEB. SEB operators were asked to increase the amount of SEBs in the future by 59 % of the users. Therefore, there would be a huge market demand for SEB (Leadleo, 2021). Fig. 8 shows the number of SEBs in China was about 2.58 million in 2021. It resulted in a maximum GHG reduction of 419.157 kt CO_2-eq according to the life cycle GHG emission accounting method in 2021. It is expected that the number of SEBs in China will increase rapidly in the next five years. In 2025, the number of SEBs in China will be expected to reach 7.945 million (Iimedia, 2020). These SEBs could achieve a maximum GHG reduction of 1290.777 kt CO_2-eq in 2025.

5. Discussions

This paper innovatively calculates the GHG emissions of SEB from production, operation and dispatch to disposal, and calculates the GHG emission reduction in the use stage. It was found that SEBs are green products in the life cycle. Scenario 1 is the one with the best environmental benefits. The production stage has the largest impact on GHG emissions at about 62 %, and the operation and dispatch stage accounts for about 30 % of total GHG emissions in scenario 1. 98 % of energy consumption during the production stage was consumed by GHG emissions from raw materials. By further analyzing the materials input at the production stage, we find that the use of aluminum alloys has caused the maximum negative environmental impact. This is mainly because aluminum production mainly relies on the electrolytic method, which consumes a lot of electricity. If more environmentally friendly materials, such as composite materials, are used in production instead of aluminum alloys, the GHG emissions of SEBs will be greatly reduced. Although we collected all kinds of components of SEB and made various assumptions, it may cause some errors in the calculation results due to the use of different types of materials in the production stage, such as glass fibers, photographic materials, and copper in electronic systems. In addition, since the content of a single emission factor database could not meet the requirements of the paper, we had to refer to several databases and some research results. Although we have carefully identified emission factors for each material, the multiple sources of emission factors may introduce some uncertainty in the results. The replacement rate of SEBs for high-emission transportation modes determines the emission reduction effect in the use stage. Due to the lack of a complete and effective recycling system, we were unable to assess the GHG emission reduction generated during the recycling process of SEB, which is crucial to enhance the environmental benefits of SEB. Therefore, it can be considered that optimizing SEB production, encouraging green travel, and establishing a recycling system may be the future direction of green development of SEB. We provide the following solutions to the practical issues of electric bicycle sharing in energy savings and GHG reduction.

Table 6
Total net GHG reduction in different disposal scenarios.

Scenario	Production stage	Operation & dispatch stage	Disposal stage	Use stage	Unit	Net GHG reduction (kg/SEB)
H1	379.6173	183.4663	0.6385*	1049.8374*	kg CO_2-eq	487.3923
H2	379.6173	183.4663	48.3133	1049.8374*	kg CO_2-eq	433.9215

Notes:
* Is for GHG reduction.

5.1. Promoting SEB to replace high-emission transportation modes

It was found that SEB had a substitution effect on other transportation modes and could result in an effective reduction of GHG emissions. However, SEB mainly replaced low-GHG emissions or zero-GHG emissions transportation modes such as walking, buses, and shared bicycles, which accounted for 69.5 %. The replacement rate for high-GHG emissions transportation modes such as private cars and taxis was low. Therefore, promoting SEB to replace high-GHG emission transportation modes is one of the important ways to reduce more GHG emissions. It was shown that the average single trip distance of the SEB was 3.06 km. The travel distance of high-GHG emissions transportation modes is mainly concentrated in 8–20 km. Increasing the trip distance of SEB is the key to replacing high-emission transportation modes. Cities with SEBs can plan to develop slow traffic systems, set up dedicated lanes for SEBs, improve road signs and traffic markings, and remove traffic barriers. It is necessary to plan the parking areas for SEBs to solve the problem of hindering traffic by parking indiscriminately. Finally, green travel should be actively promoted. SEB companies should improve their operation level to increase their market share of SEBs. These measures will ensure that the SEBs are fully charged and trouble-free to improve the user experience.

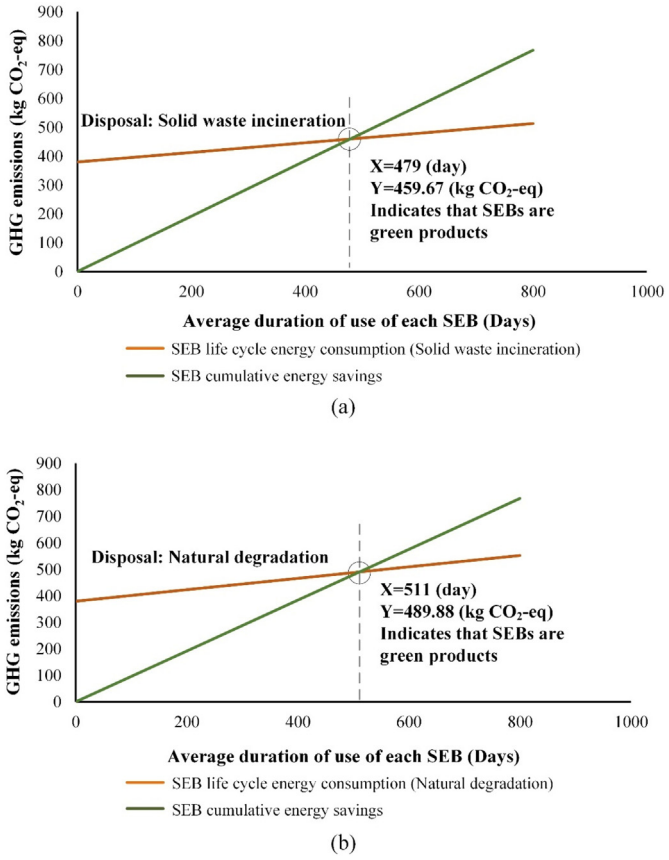


Fig. 6. GHG emissions threshold in days. (a) In the recycling incineration scenario (H1); (b) In the recycling degradation scenario (H2).

5.2. Determining the number of SEBs according to the capacity of the market

Currently, there are more than ten kinds of SEBs under eight brands in the China market where competition among SEB operators is fierce. SEB operators are trying to gain customer loyalty by increasing the number of SEBs to increase their market share and gain sufficient profits. However, SEB operators only pay attention to economic benefits in the process of attracting users. The increase in the number of SEBs will lead to a gradual increase in the ratio of empty SEBs, which will affect GHG emission thresholds of SEB and reduce the environmental benefits. If there is no scientific plan for the efficiency and amount of SEBs, it will be difficult for SEB to achieve energy conservation and GHG reduction. The government can advocate SEB companies to account for GHG emission reductions. According to the market demand and urban traffic conditions, it is necessary to promote a quota for the number of SEBs to achieve control of the number of SEBs of various SEB companies. SEB companies should change their business strategies and focus on how to improve the quality of service, improve energy saving and emission reduction, continuously improve the user experience and enhance the social image.

5.3. Improving production and maintenance to increase the service life of the SEB

GHG emissions produced in the production stage were the main parts of the whole GHG emissions in the life cycle. It included the production and processing of materials. The production and operation stage emissions accounted for 62 % and 30 % of total GHG emissions respectively. 98 % of energy consumption during the production stage was consumed by GHG emissions from raw materials. As a result, the primary source of GHG emissions from SEBs was energy consumption of such materials. Environmental benefits of SEB could be projected to increase significantly with better energy consumption management in the production stage. In particular, the

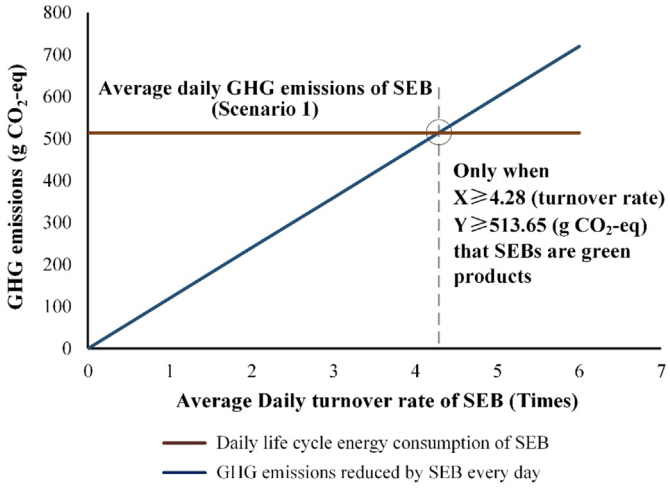


Fig. 7. GHG emissions threshold in turnover rates per day.

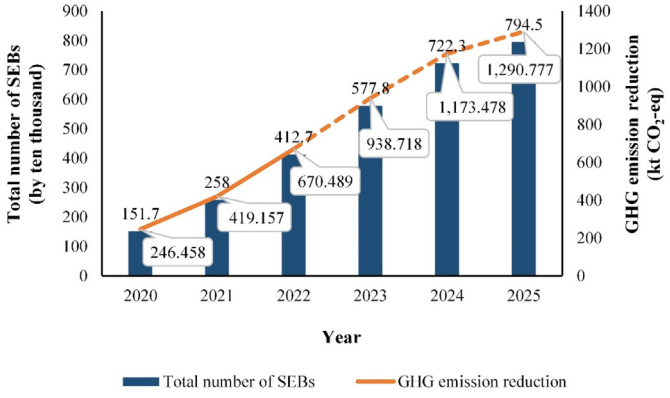


Fig. 8. the number of SEBs and GHG reduction in China.

strict management of the production process, optimization of the production process, and advancement of production technologies are the keys to achieving GHG reduction. The actual service life of the SEB is defined by the least durable component. Therefore, the correlation between environmental effect and material durability becomes one of the principles of material selection. Manufacturers can design SEB with more easily disassembled maintenance modules that are easier to repair through the assembly line to reduce maintenance cost of SEB. SEB operators should distribute each SEB as fairly as possible during the life cycle to avoid damage to susceptible components. It will increase the service life of SEB. Overall, identifying an appropriate solution is critical to sustainable development of SEB.

5.4. Improving the disposal stage of the SEB to create a circular industrial chain

Electric bicycle-sharing should be considered a green product. A comprehensive and efficient recycling system for SEB is absent currently. Private institutions remain essential to the recycling of the SEB. The lack of a consistent recycling system results in inefficiency and waste. As a result, the recycling market for SEB is not active (MOC, 2020) and, the lack of a complete recycling system made it impossible to calculate the deduction effect of the recycling process in this LCA calculation. More efficient electric bicycle sharing recycling businesses will considerably reduce environmental consequences at the production stage from a life cycle perspective. The government should improve the policy system design, such as the extension of producer responsibility system and recovery compensation system. It is necessary to promote cooperation between recycling companies and SEB companies, which is conducive to the fixed-point recovery and centralized disposal of abandoned SEBs. It is effective for recycling SEBs by establishing new recycling processing models and recycling platforms, which need the support of information technology, such as big data, blockchain, and artificial intelligence. In addition, we should also promote the business linkage of production enterprises, operating enterprises, recycling enterprises, and waste treatment enterprises based on the life cycle to build a SEB recycling system. Finally, waste battery treatment technology should be focused on to improve the recycling level and pollution prevention capacity of the waste battery recycling industry.

6. Conclusion

Transportation is the backbone of the national economy and society in China. GHG emission reduction in transportation has become an important area in the country. Shared transportation is a new way to reduce emissions. SEB has become a new green shared transportation mode after public bicycles and shared bicycles.

The life cycle analysis approach was used in this study to analyze the factors and main parameters that affect the energy consumption of SEB. The life cycle GHG emissions accounting model for SEB was established based on the obtained factors and parameters. Meituan SEB, the most common type of electric bicycle sharing, was chosen as the object of the current research. This study evaluated resource consumption and GHG emissions in each stage of production, operation, and disposal of the SEB. The differences in GHG emissions between the recycling incineration scenario and recycling degradation scenario were compared in the disposal stage. Moreover, net GHG emissions in the whole life cycle of the SEB were calculated separately under different disposal scenarios. GHG emission thresholds of the SEB were analyzed in terms of operating days and daily turnover rates. Finally, we found that the GHG emissions in the production and operation stages were 379.6173 kg CO_2-eq and 183.4663 kg CO_2-eq respectively. GHG emission reduction in the use stage was 1049.8374 kg CO_2-eq in the whole life cycle. GHG emissions differed greatly between the two different disposal scenarios. GHG reductions of SEB were 0.6385 kg CO_2-eq in the recycling incineration scenario. GHG emissions of SEB were 52.8323 kg CO_2-eq in the recycling degradation scenario. Therefore, net GHG reductions in the life cycle of the SEB were 487.3923 kg CO_2-eq (Scenario 1) and 433.9215 kg CO_2-eq (Scenario 2). It was indicated that SEB might be a green product. If a SEB was deposited in a landfill, it would take 48 years to degrade. The final calculation showed that if the daily average turnover rates were less than 4 and the operating days were less than 479, the SEB would not be able to achieve the goal of GHG reduction in the whole life cycle.

Although this study comprehensively evaluates the GHG emission impact of each stage of SEB, several important future research topics could be studied to enhance the existing research, such as optimizing SEB production, encouraging green travel, and establishing a recycling system. Shared economy is very important to the sustainable development of society. In the future, the role of many products in the shared economy, such as transportation, logistics and express delivery, and life service, still needs to be explored more thoroughly to achieve the sustainable development of society.

CRediT authorship contribution statement

Methodology, Z.Z.; data curation, Z.Z.; supervision, C.L.; writing—original draft, Z.Z. and C.L.; writing—review and editing, Z.Z. and C.L. All authors have read and agreed to the published version of the manuscript.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Appendix

Table A
Meaning of the nomenclatures in the paper.

Nomenclature	Meaning
EC_n	The *Nth* energy consumption of producing the SEB
EF_n	The *Nth* GHG emission factor of producing the SEB
EF_{SEB}	The GHG emission factor of the SEB
DIS_{avg}	Average travel distance of the SEB each time
TR_i	The replacement rate of the SEB to travel mode *i*
TF_i	The GHG emission factor of travel mode *i*
H	Average power consumption per kilometer of the SEB
EF_{day}	Average GHG emissions produced by a SEB are rebalanced every day
T_l	Time length of the life cycle of the SEB
ME_m	Repair probability of the *mth* component
MSW_i	Weight of municipal solid waste of type *i* of component of the SEB
EF_i	The GHG emission factor of incinerating type *i* of component of the SEB

(continued on next page)

Table A (*continued*)

Nomenclature	Meaning
MSW_j	Weight of the *jth* pollutant
EF_j	The GHG emission factor of the *jth* pollutant
$NDWT_c$	Weighted coefficient of natural degradation time
X_i	The proportion of emissions to total emissions in the *ith* year
T_{ET}	Average existing time for the total GHG emissions released by the SEB in the atmosphere
PPE	GHG emissions in the production process of the SEB
UPE	GHG emissions in the use stage of the SEB
OPE	GHG emissions in the operation and dispatch process of the SEB
RPE	GHG emissions in the rebalancing section of the SEB
MPE	GHG emissions in the maintenance section of the SEB
DPE_{H1}	GHG emissions of scenario 1 (recycling incineration scenario)
DPE_{H2}	GHG emissions of scenario 2 (recycling degradation scenario)
LPE	Net GHG emissions during the whole life cycle of the SEB

Table B
Summary of related works and data source.

Source	Content	Use for reference
SPIR (2020)	Time length of the life cycle of the SEB	Calculation of GHG emissions in use stage and operation stage
CCG (2021)	The average GHG emission factor of the grid in China in 2020	Calculation of GHG emissions in the use stage
SUNRA G-force electric bicycle store	Energy consumption during the production of the SEB	Calculation of GHG emissions in the production stage
Meituan	Energy consumption during the operation of the SEB	Calculation of GHG emissions in use stage and operation stage
CPCD database EFDB database of IPCC Carbonstop Hirano and Fujita (2016)	GHG emission factors during the production and disposal stage of the SEB	Calculation of GHG emissions in the production stage and disposal stage
Google Maps	GHG emission factors during disposal of the SEB	Calculation of GHG emissions in the disposal stage
The questionnaire designed by the research team	The distance for vehicles to transport the SEB in different stages of the whole life cycle	Calculation of GHG emissions in the production stage, operation stage, and disposal stage
GB50157-2013 GB17761-2018 GB19578-2021 CPCD database Didi (2020)	The replacement rate of the SEB with other transportation modes	Calculation of GHG emissions in the use stage
GBT36944-2018	GHG emission factors of different transportation modes at the use phase	Calculation of GHG emissions in the use stage
GB50688-2011	Charging efficiency of 48 V 20 Ah battery	Calculation of GHG emissions in the use stage
CJJ17-2004	Average road speed in class C and D cities in China	Calculation of GHG emissions in the operation stage
CJJ/T107-2019	Distance between the landfill site and the urban built-up area	Calculation of GHG emissions in the disposal stage
	The maximum design life of the landfill site	Calculation of GHG emissions in the disposal stage

Table C
Data description of operation and dispatch stage.

Description	Value	Unit	Reference
Emission factor for small truck	0.083	kg CO_2-eq/(t·km)	Carbonstop
Emission factor for electric three-wheeler	0.081	kg CO_2-eq/km	CPCD database
Number of SEBs dispatched per day	120	–	Interviews with dispatchers of Meituan SEB
Number of daily dispatch of SEBs	6	–	Interview with dispatcher of Meituan SEB
Time consumption per dispatch of SEBs	1	Hour	Interviews with dispatchers of Meituan SEB
Daily battery replacement time for SEBs	14	Hour	Interviews with staff from Meituan SEB
Number of batteries replaced per day	300	–	Interviews with staff from Meituan SEB
The proportion of SEBs requiring battery replacement	25	%	The hypothesis in the paper
Average road speed in class C and D cities in China	30	km/h	GB 50688-2011

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